



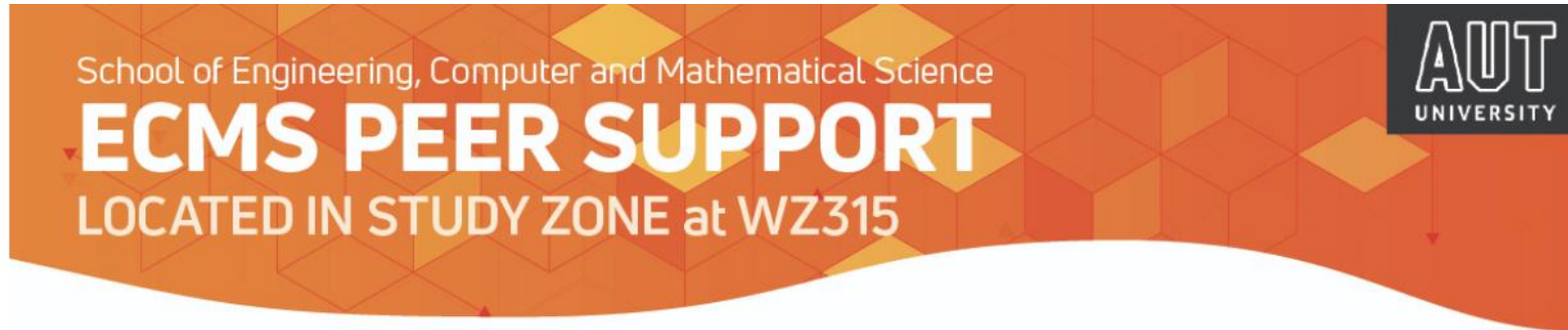
MATH503



Statistical measures

# Housekeeping

- WZ Level 3 Study Space (extra support)



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**UNDERSTANDING** academic concepts

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# Housekeeping

➤ MATH503 Manual

Revise Chapter 3, Sections 3.1 and 3.2:

Section 3.1 – Measures of central Tendency

Section 3.2 – Measures of Spread (variability)

> R Studio !!



# Myimaths details

- Individual portal username and password
- List allocated tasks
- 25% final grade

# Discrete or Continuous Data

## > Discrete and Continuous Data

Statistics involves collecting, organising and displaying the data obtained from samples. Data are the pieces of information collected in the sample or survey.

**There are 2 types of data: Discrete and Continuous**

> **Discrete data** arise from counting and usually involve whole numbers.

**Example: How many people took the bus to University today?**

> **Continuous data** arises from measuring and can be any value

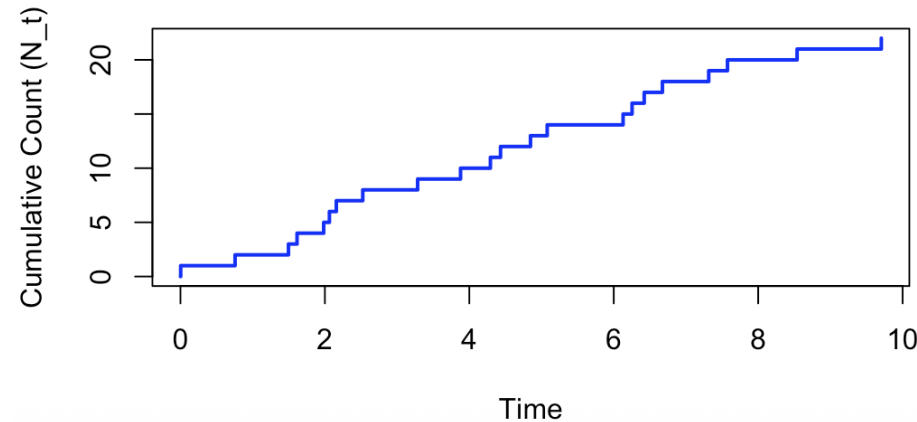
**Examples: How long did it take you to travel to University today?**

**How long from when you left home till you arrived on campus?**

# Discrete or Continuous Data : other examples

- **Discrete data** arise from counting and usually involve whole numbers.

The signal that accumulates in each pixel of a CCD camera during exposure is due to Photons (discrete quanta of light) arriving from a source. You can not have a fraction of a photon, the signal count increases by integer amounts during an exposure

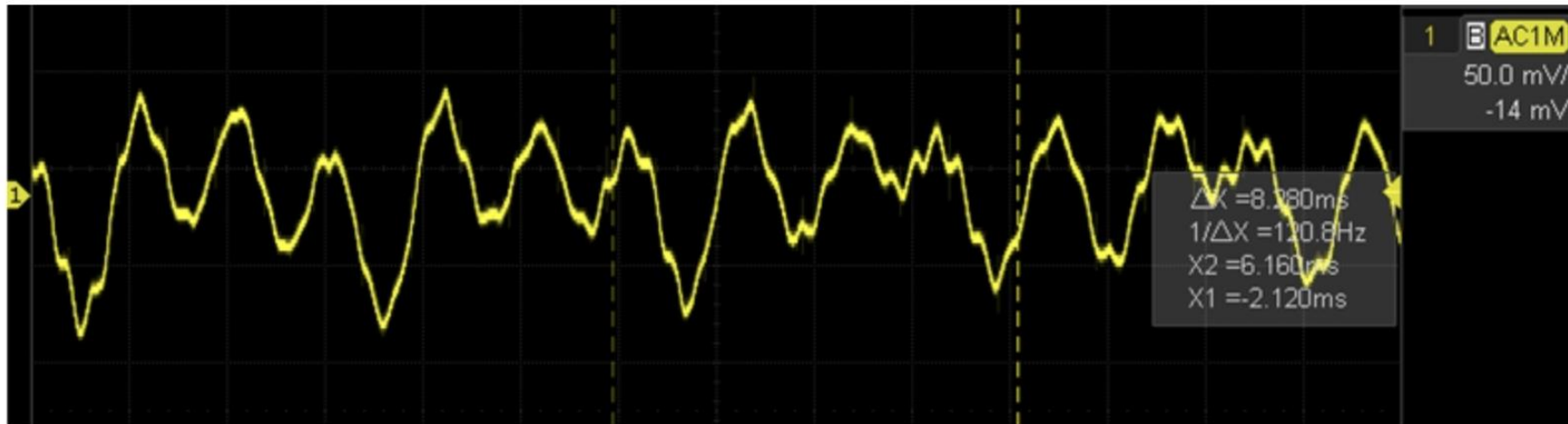




# Discrete or Continuous Data : other examples

- **Continuous data** typically arises from measuring analogue signals and can be any value

The voltage measured across the terminals of a microphone as a function of time

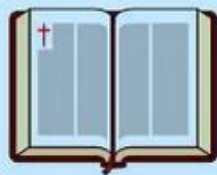


# Descriptive statistics

Descriptive statistics is a means of describing features of a data set by generating summaries about data samples.

## EXAMPLE 1:

The average test score for the students in a class, to give a descriptive sense of the typical scores.



## EXAMPLE 2:

According to Consumer Reports, there were 2.5 problems per one copying machines reported during 2009.





# Descriptive statistics

- **Measures of central tendency**

Averages: mean, median, mode

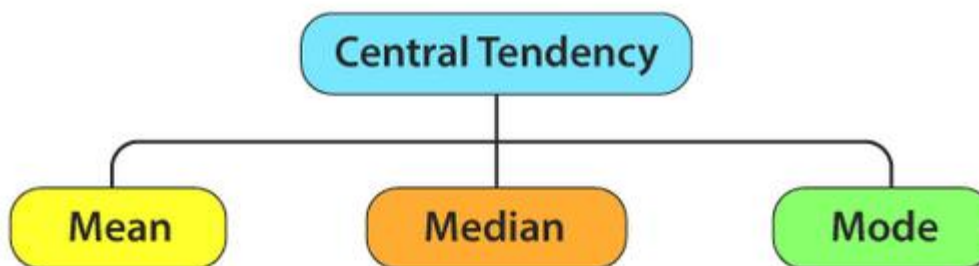
- **Measures of spread (variability)**

Spread of population: variance, standard deviation, range, maximum, minimum, lower quartile, upper quartile, interquartile range

# Measures of central tendency (Averages)

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.

## CENTRAL TENDENCY



**The mean, median and mode are all 'measures of central tendency'**

Under different conditions, some measures of central tendency become more appropriate to use than others.

The **mean** is the most popular and well-known measure of central tendency.

**The mean is equal to the sum of all the values in the data set divided by the number of values in the data set.**

If we have  $n$  values in a data set and they have values  $x_1, x_2, \dots, x_n$  the sample mean, usually denoted by  $\bar{x}$  is:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

This formula is written in a slightly different manner using the Greek capitol letter,  $\Sigma$ , pronounced "sigma", which means "sum of...":

$$\bar{x} = \frac{\sum x_i}{n}$$

# Example

Example: running times for an algorithm

4.3s, 4.2s, 4.7s, 4.3s, 4.1s, 4.2s, 4.3s, 4.25s, 4.2s, 4.4s

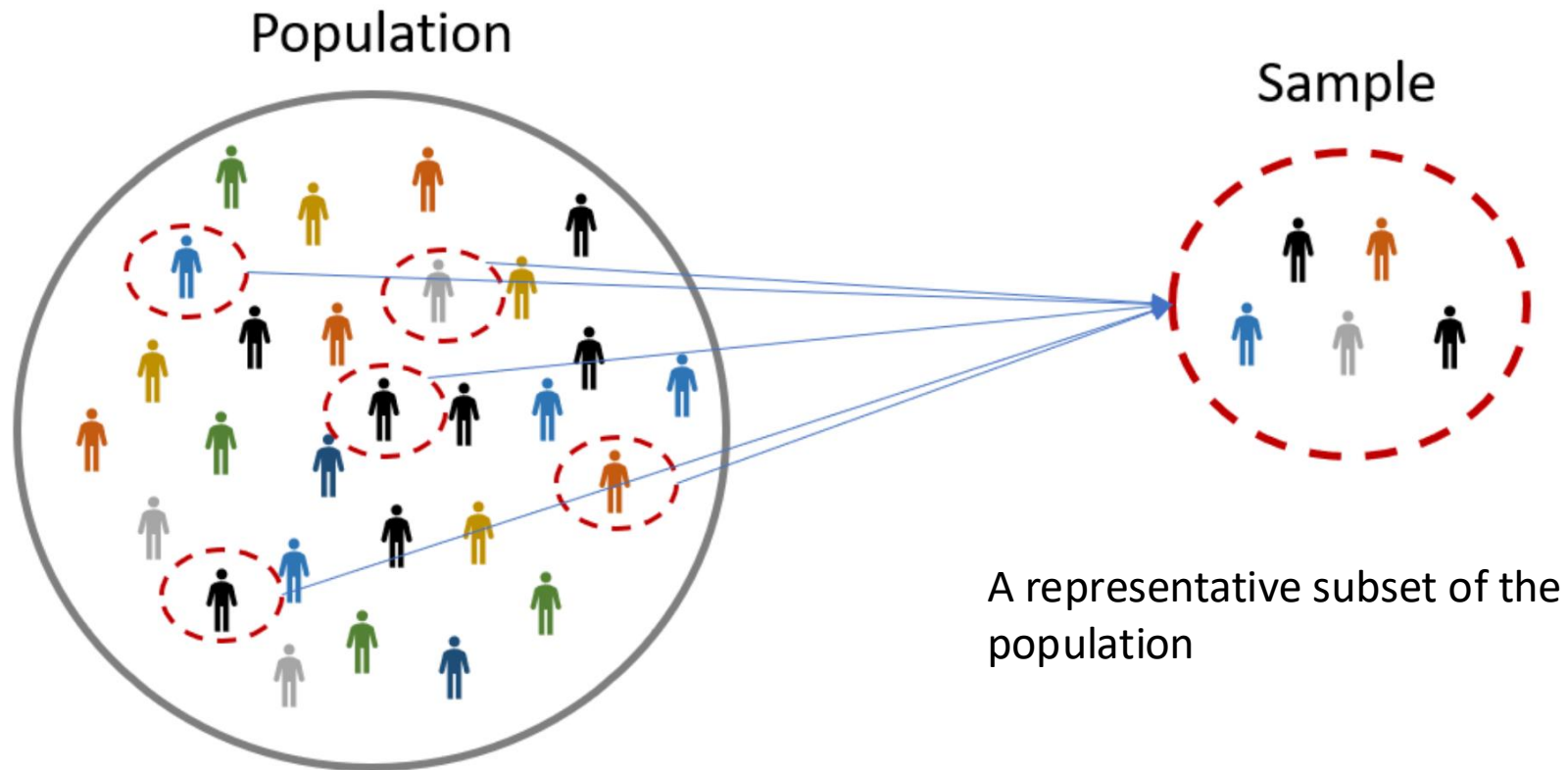
Calculate the mean running time

$$\bar{x} = \frac{4.3 + 4.2 + 4.7 + 4.3 + 4.1 + 4.2 + 4.3 + 4.25 + 4.2 + 4.4}{10}$$

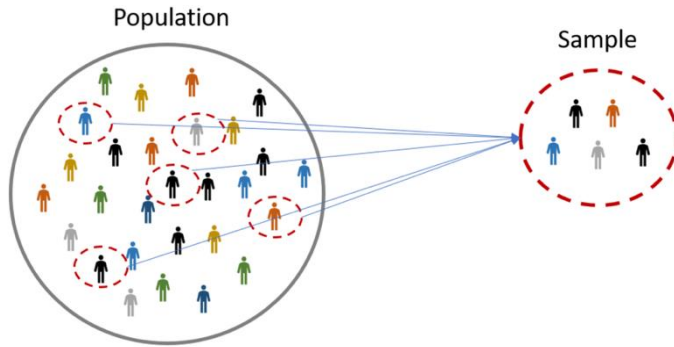
$$\Rightarrow \bar{x} = \frac{42.95}{10} = 4.295 \approx 4.3 \quad (\text{to 2 s.f.})$$

# Population vs Sample

**In statistics sample and population have different meanings.**



# Population vs Sample



- We typically want to calculate a statistic that tells us about the expected behaviour/state/... of a complete population/system; e.g. who the majority of people are going to vote for in an election.
- The best way to do this would be to ask every eligible voter, i.e. to question/sample the entire population.
- This is frequently impractical, there are too many members of the population to do this; e.g. there were 3,592,978 registered voters eligible for the 2026 NZ General election as of 19<sup>th</sup> March 2026 ( see <https://elections.nz/stats-and-research/enrolment-statistics>)
- Next best approach is to survey a subset of manageable size, the 'Sample'.
- We infer the behaviour of the population from the responses of the sample.
- Need to be careful so the sample returns a result that is representative of the larger population. This can go wrong!
- See <https://www.youtube.com/watch?v=bkzS-99So2U&t=1007s> for accepted ways to choose a sample.



| Population Mean                                                                              | Sample Mean                                                                                  |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| $\mu = \frac{\sum_{i=1}^N x_i}{N}$ <p><math>N</math> = number of items in the population</p> | $\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$ <p><math>n</math> = number of items in the sample</p> |

> The population mean is usually denoted by  $\mu$

**In statistics sample and population have different meanings.**

EXAMPLE: This semester's MATH503 students- The population is the total number of students enrolled in this course (249 Students S1 2026). A sample of this population would be **a randomly selected group of 30 students** enrolled in this course.

The sample is only part, but correctly selected, of the population, hence why we use  $\mu$  to denote the population mean and  $\bar{x}$  to denote the sample mean.

Population for S1 2022 – 299 students

Population for S1 2024 – 377 students

Population for S1 2025 – 307 students

Population for S1 2026 – 280 students



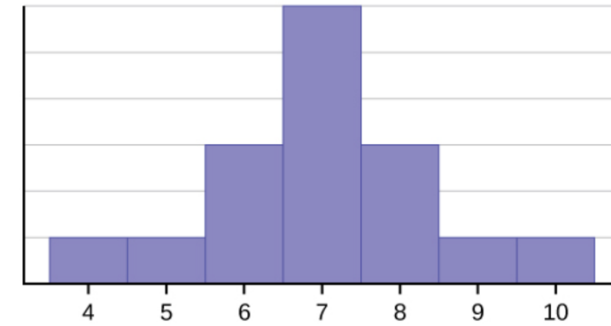
# Mean as a balance point

<https://teaching.statistics-is-awesome.org/helping-students-to-estimate-mean-and-standard-deviation/>

“exploring the balance point”

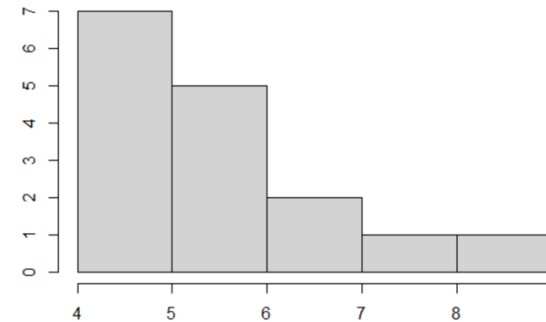
Consider the following data set:  
4,5,6,6,6,7,7,7,7,7,7,8,8,8,9,10

$$\bar{x} = 7$$



Consider the following data set:  
4,4,5,5,5,5,5,6,6,6,6,6,7,7,8,9

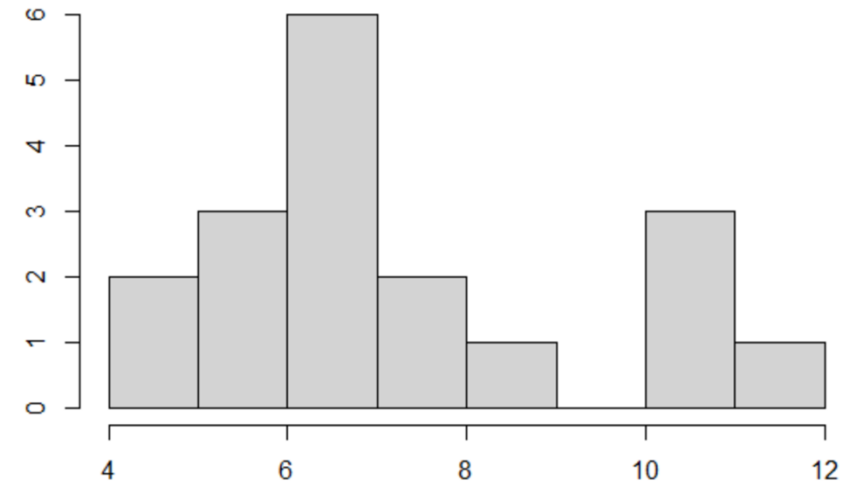
$$\bar{x} = 5.875$$



Consider the following data set:

4,5,6,6,6,7,7,7,7,7,7,8,8,9,11,11,11,12

$$\bar{x} = 7.72222..$$



## Example of when NOT to use the mean (skewed distribution):

|        |     |     |     |     |     |     |     |     |     |      |      |
|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| Staff  | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10   | 11   |
| Salary | 45k | 48k | 56k | 54k | 55k | 55k | 46k | 32k | 47k | 120k | 135k |

The mean salary for these eleven staff is \$63k.

**In this case, is the mean the best way to accurately reflect the typical salary of a worker?**

**Most workers have salaries in the \$45k to 56k range.**

**The mean is being 'skewed' (affected) by the two large salaries.**

>> In this situation, we would like to have a better measure of central tendency.

The **median** would be a better measure of central tendency in this situation.

# Median

**The median is the middle score for a set of data that has been arranged in order of magnitude.**

> To find the median, first order the data in order of magnitude, generally from smallest to largest. Then find the middle score

For example: 3sec, 4sec, 6sec, 8sec, 11sec

We can see 6 is the middle value, therefore 6 is the median

3, 4, 6, 8, 11

**Interpretation: 50% of the times recorded are below 6 seconds !**

# Median

Example II: 2seconds, 3seconds, 4sec, 5sec, 7sec, 9sec

For this data see the middle value lies between 4 and 5, therefore the median is 4.5

Calculation:  $\frac{(4+5)}{2} = 4.5$

2, 3, 4 | 5, 7, 9

Interpretation: 50% of the times recorded are below 4.5 seconds !



# The median is always representative of the middle data and is less affected extreme data (outliers).

For example:

> Suppose we use the salary data:

45k, 48k, 56k, 54k, 55k, 55k, 46k, 32k, 47k, 120k, 135k

> First we order the data (generally smallest to largest)

32k, 45k, 46k, 47k, 48k, 54k, 55k, 55k, 56k, 120k, 135k

Now find the middle value

54k (mean was 63k)





## In summary: median

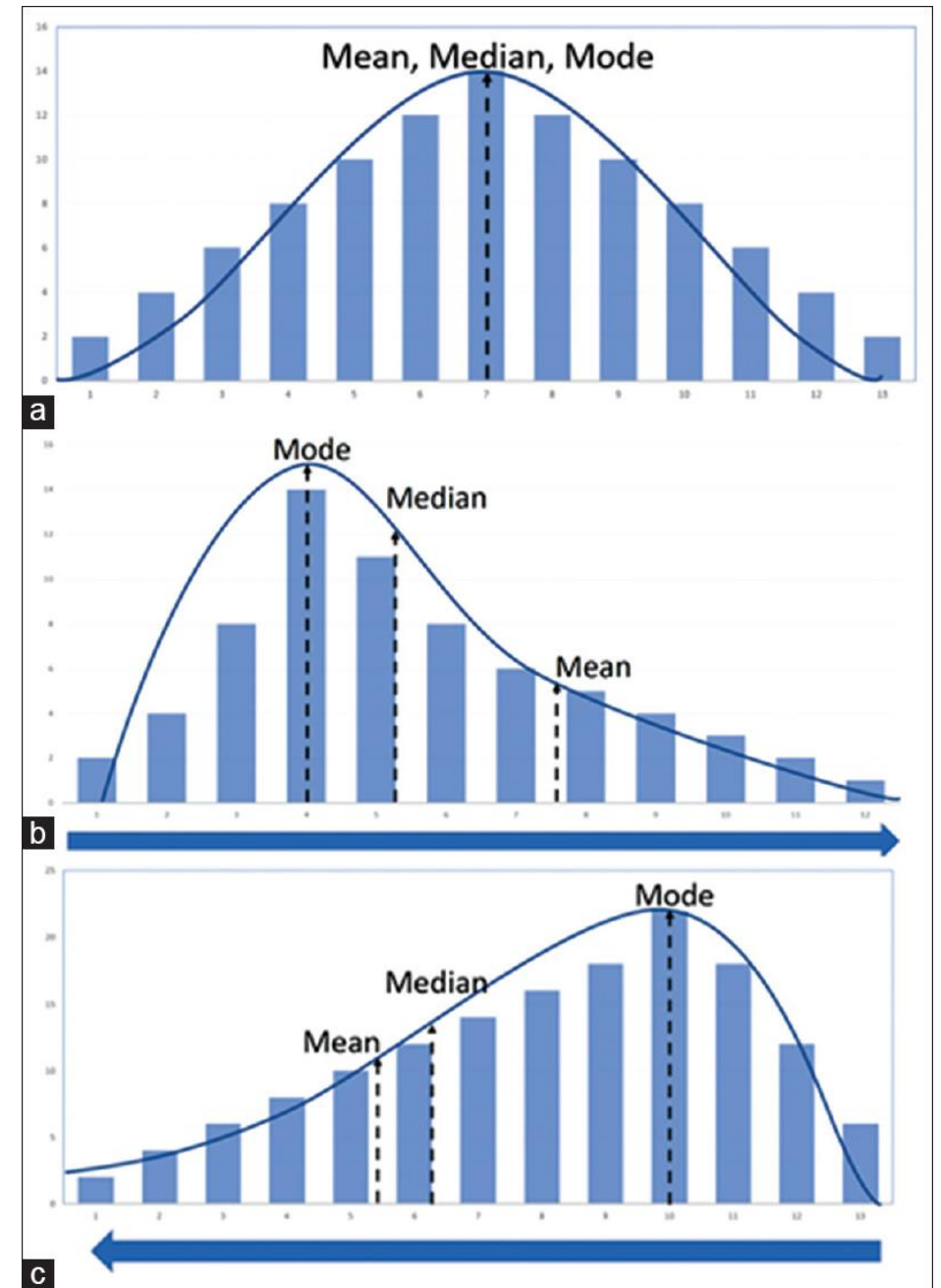
- The median is less affected by outliers and skewed data
- Always representative of middle data
- Data must be in order
- Not swayed by extreme values
- Good to use when outliers are present in the data

# Skewed distributions

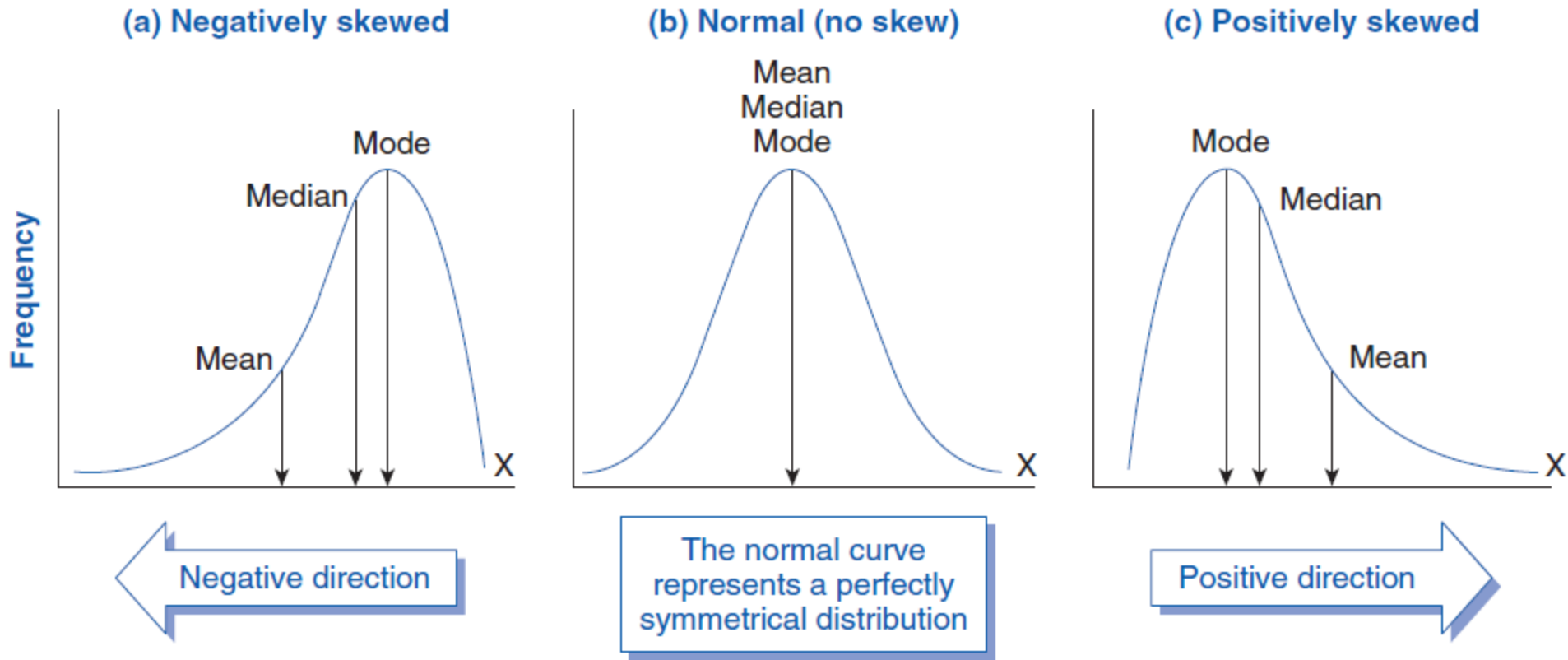
Normal (no skew)

Positively skewed

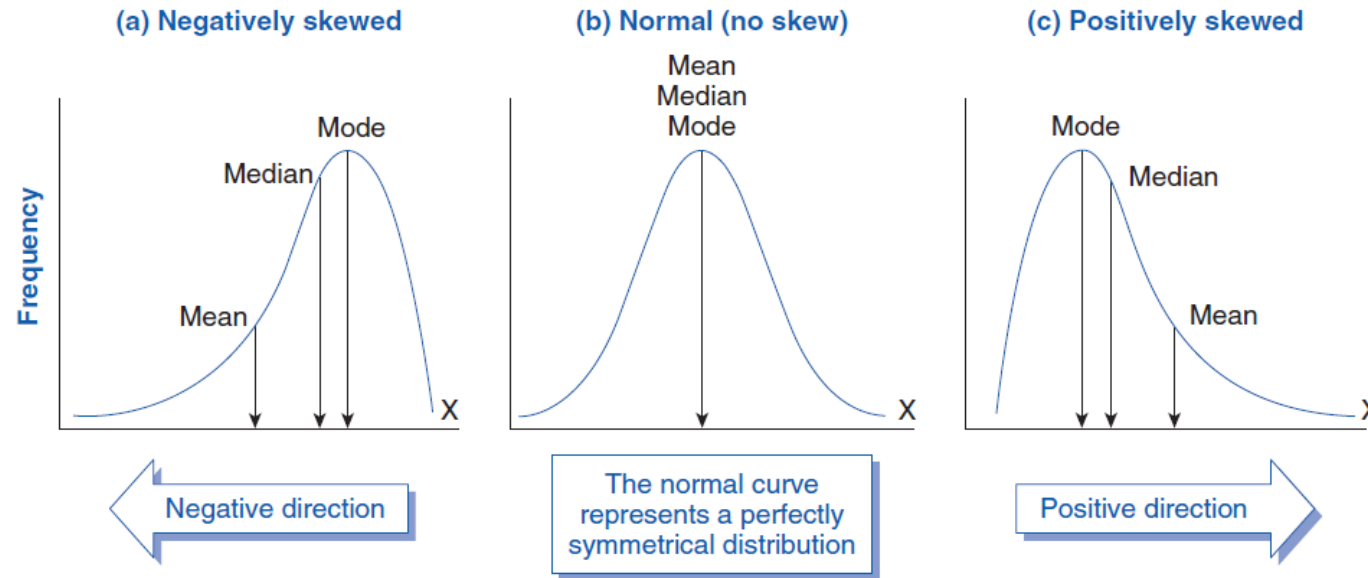
Negatively skewed



# Skewed distributions



# Skewed distributions



**As the data becomes skewed the mean loses its ability to provide the best central location for the data because the skewed data is dragging it away from the typical value.**

## Summary: sample mean (left) vs population mean (right)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} \quad \text{or} \quad \mu = \frac{x_1 + x_2 + \dots + x_N}{N}$$

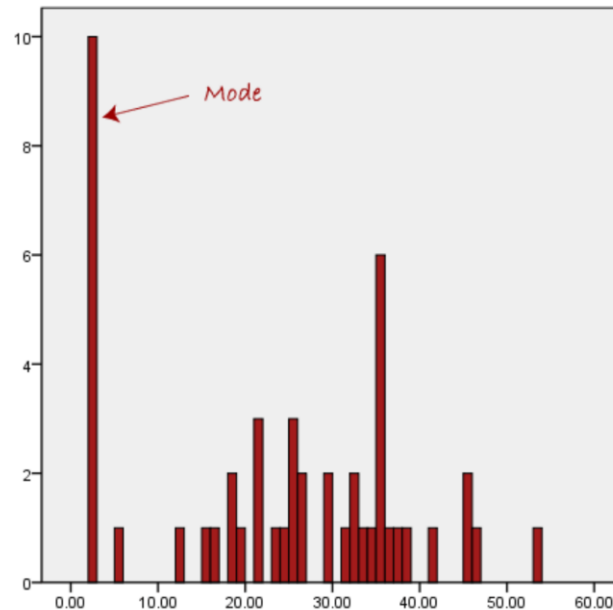
- Every number contributes to the mean
- Affected by outliers
- Good to use when data is symmetrical or does not contain any outliers.
- May not be representative of any value

# Mode

**The mode is the most common value in the data set.**

This is why the mode is very rarely used with continuous data.

Another problem with the mode is that it will not provide us with a very good measure of central tendency when the most common mark is far away from the rest of the data in the data set, as depicted in the diagram below:



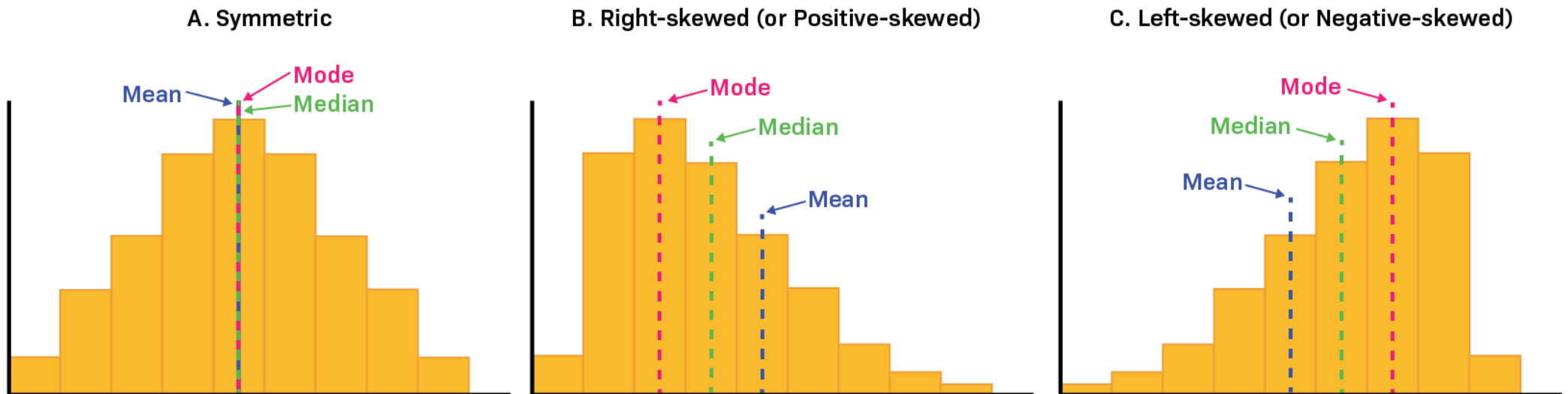
In the above diagram the mode has a value of 2. We can clearly see, however, that the mode is not representative of the data, which is mostly concentrated around the 20 to 30 value range. To use the mode to describe the central tendency of this data set would be misleading.



# Mode

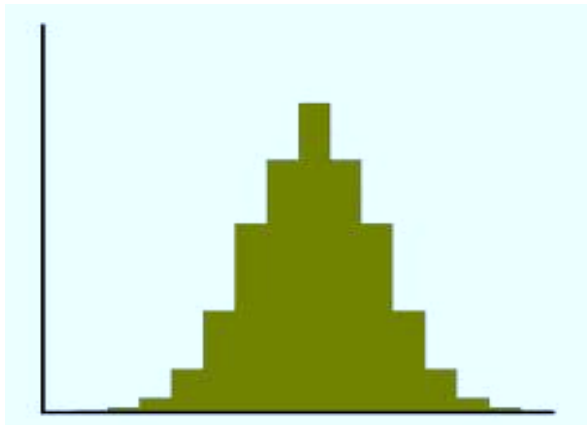
**The mode is the most common value in the data set.**

This is why the mode is very rarely used with continuous data.



# Histograms and mode

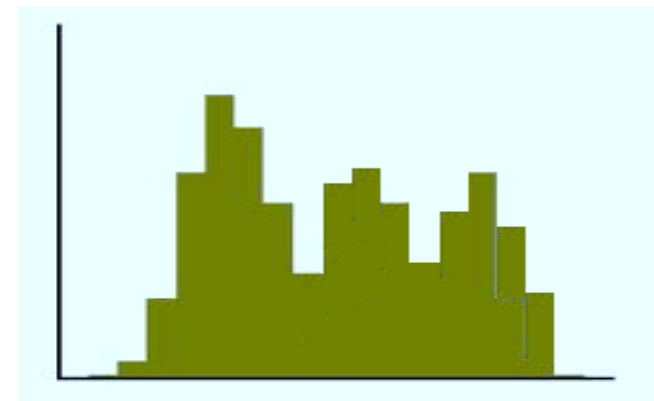
The best way to see a dataset's shape is to draw a histogram.  
We then use the mode(s) to describe the plotted data as either:



Unimodal  
Has a single prominent peak.



Bimodal  
Has two prominent peaks.



Multimodal  
When there are more than 2 peaks.



# Mean from a table

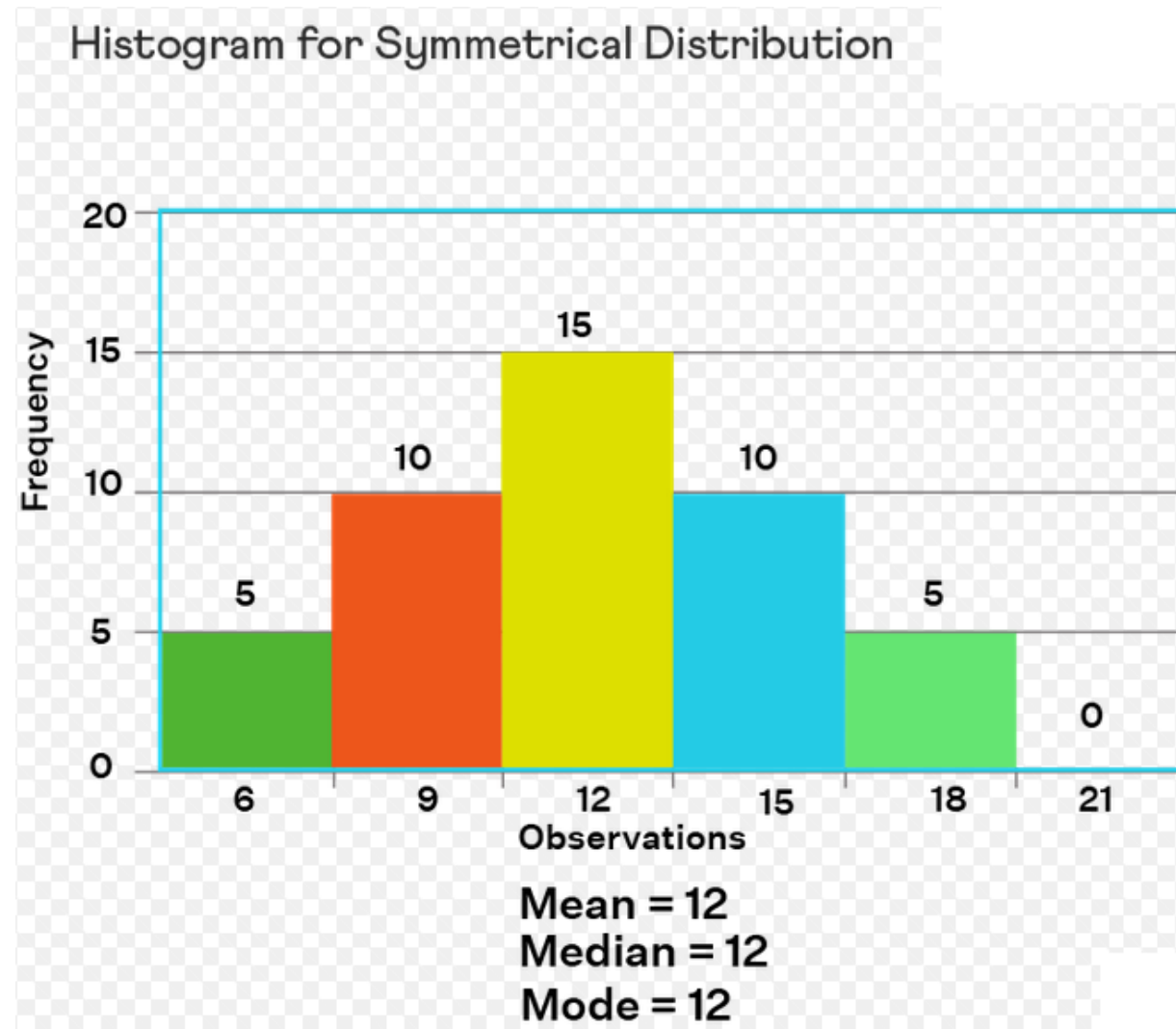
<https://app.myimaths.com/2193-lesson/mean-from-frequency-tables>

# How outliers effect centre and spread

[https://www.youtube.com/watch?v=vcbMinm\\_1Q8](https://www.youtube.com/watch?v=vcbMinm_1Q8)

The Effects of Outliers on Spread and Centre (1.5)

# Measures of central tendency



# R Statistical Software



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[CRAN](#)

**R Project**

```
Console Terminal Background Jobs
R 4.2.1 · ~/

R version 4.2.1 (2022-06-23 ucrt) -- "Funny-Looking Kid"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |
```

R is a language and environment for statistical computing and graphics.

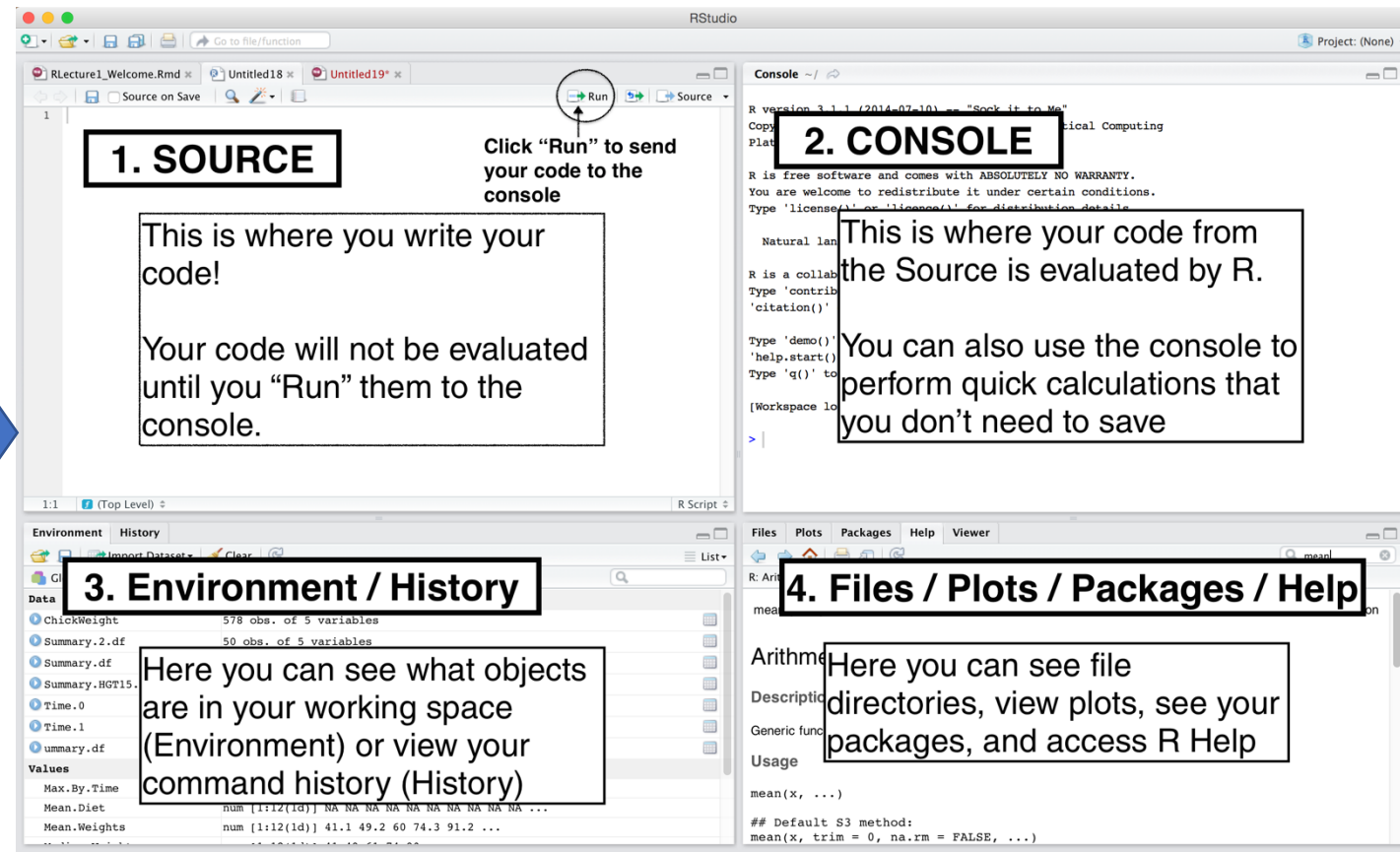
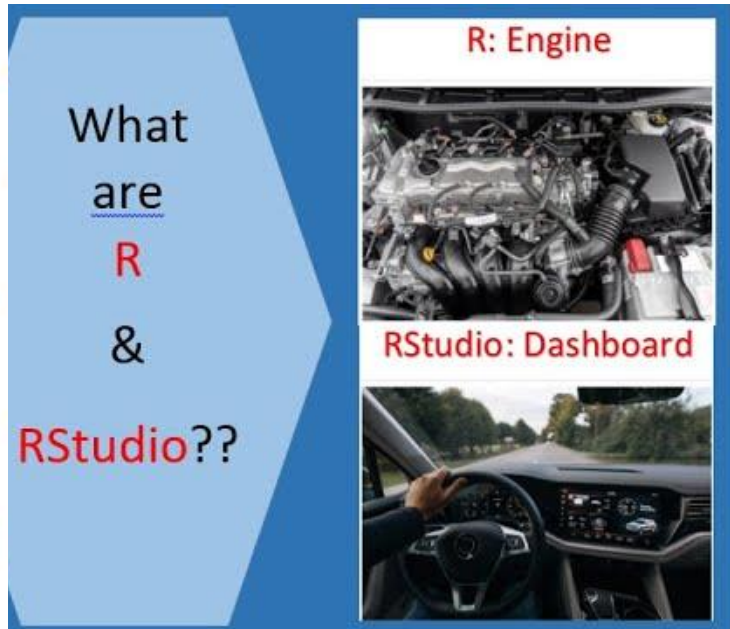
Download R:

<https://cran.r-project.org/>

R provides a wide variety of statistical (linear and nonlinear modelling, classical statistical tests, time-series analysis, classification, clustering, ...) and graphical techniques, and is highly extensible.

# R

# RStudio

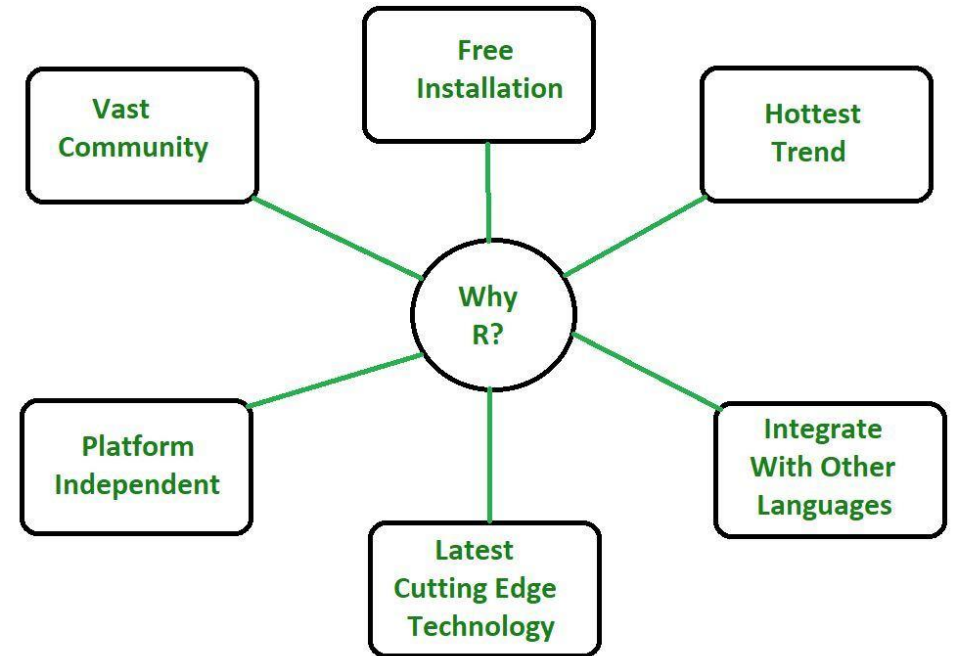


RStudio is an integrated development environment (IDE) for R.



# R and R studio

- To use RStudio at home:
- First download R from the CRAN website:
  - <https://cran.r-project.org/>
- (simply click on “download” and follow the instructions).
- Then download RStudio which makes R easier to use.
  - <https://posit.co/download/rstudio-desktop/>
- Both R and Rstudio have versions for Windows, Mac, and Linux.



# R-studio : define a data set

Use the c() operator, e.g.

➤ **data <- c(2,3,3,4,5,6,7,8)**

- c means “combine” or “concatenate”
- Creates a datastructure commonly called a vector
- Can use either arrow operator <- or = to assign data
- **Arrow operator is preferred, there are differences between the operators!**

# R-studio to calculate measures of central tendency

It is easy to calculate the mean in R using the `mean ( )` function and median with the `median ( )` function, as follows:

```
> mean(dataset)
```

```
> median(dataset)
```

R does not have a standard in-built function to calculate mode.

We will need to create a user function in order for r to calculate this for us.

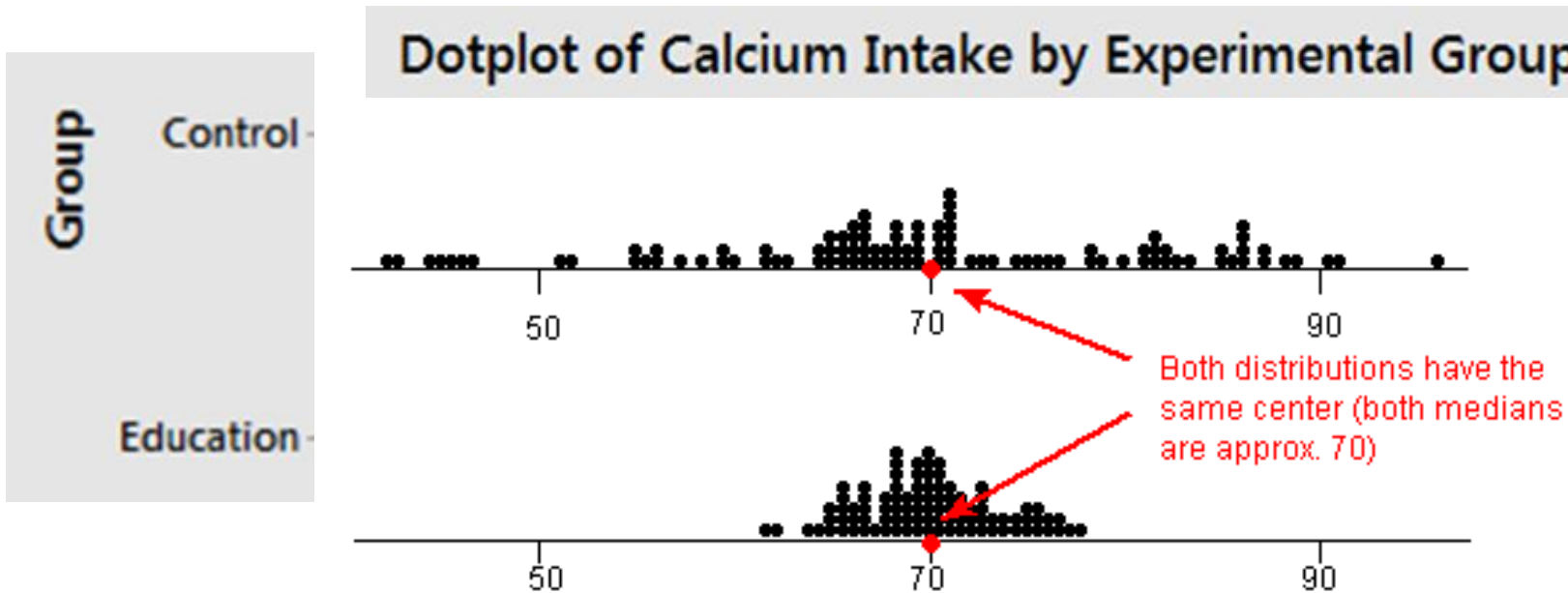
We will not be covering this in this course.

(Use Rstudio dataset AirPassengers)

# Measures of spread

A measure of spread, sometimes also called a measure of dispersion, is used to give use information about how scattered out a set of data is.

**The spread tells us about the variability in a sample or population.**

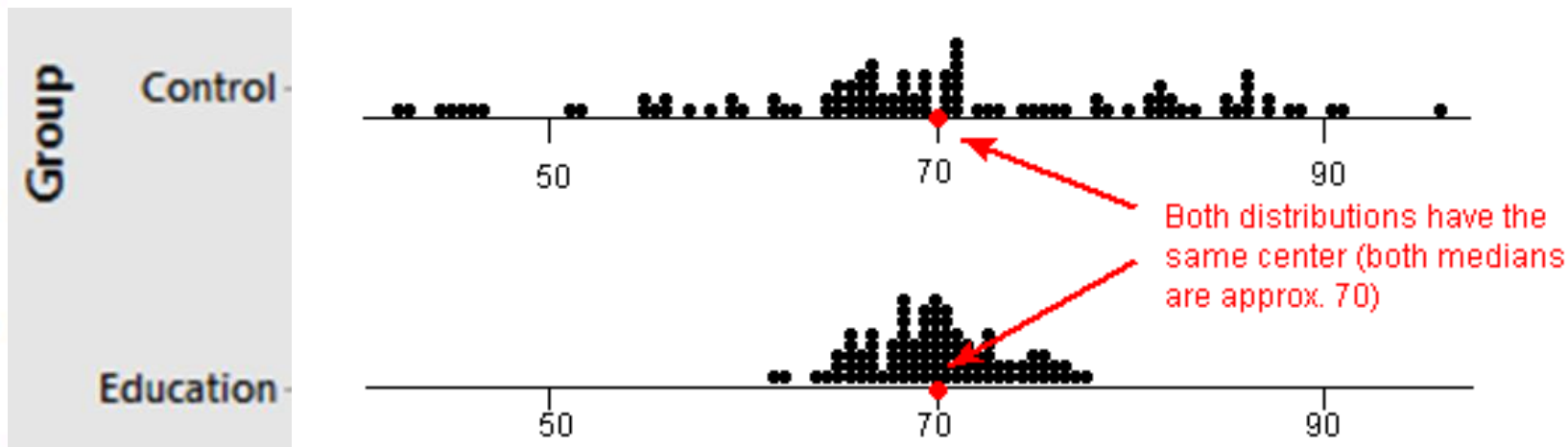


# Measures of spread

A measure of spread, sometimes also called a measure of dispersion, is used to give use information about how scattered out a set of data is.

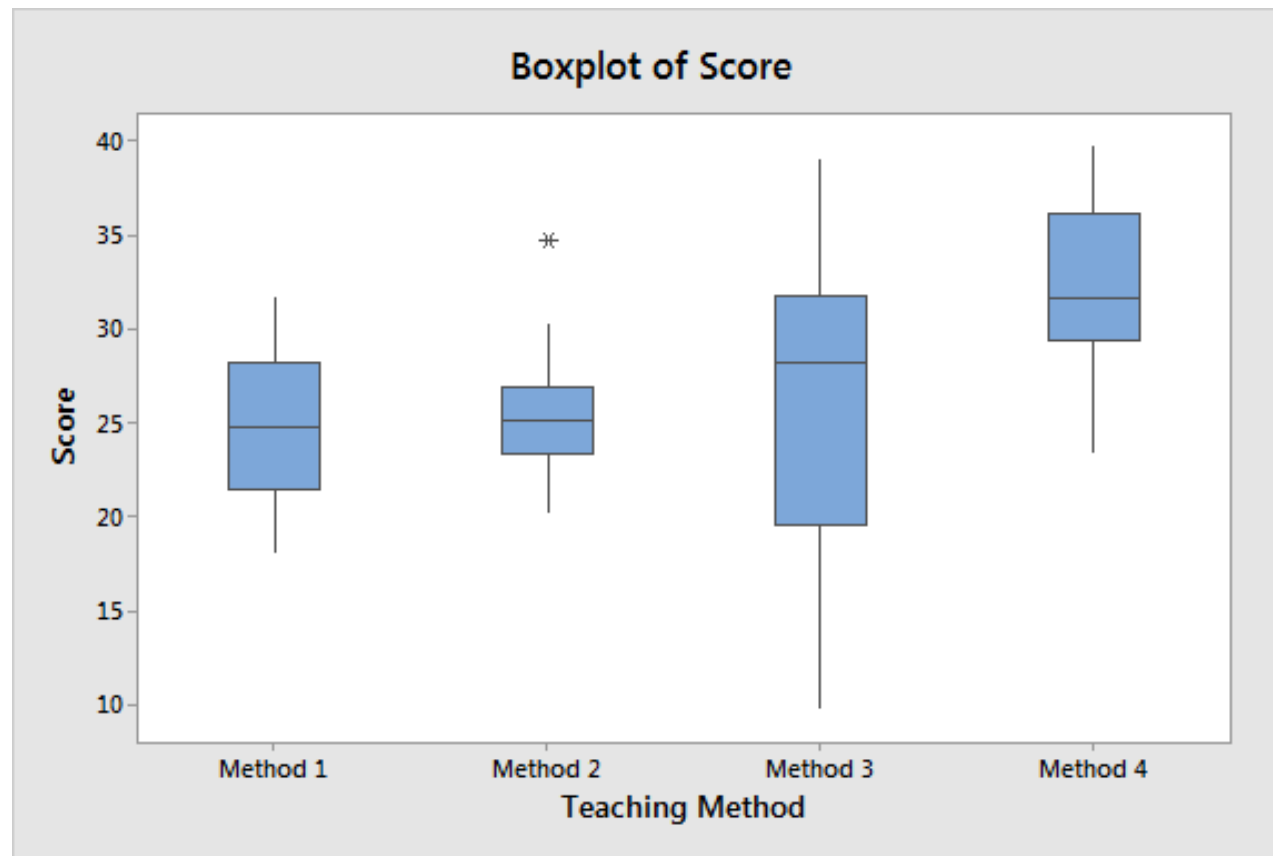
**The spread tells us about the variability in a sample or population.**

Dotplot of Calcium Intake by Experimental Group



**However, the top dotplot presents a larger variability !!**

- We need a point of comparison! A measure of spread is usually used in conjunction with a [measure of central tendency](#), such as the mean or median, to provide an overall description of a set of data.



## **Spread measures give us an idea of how well the measures of central tendency represent the data.**

For example, if the spread of values in the data set is large, the mean is not as **good** representative of the data as if the spread of data is small. This is because a large spread indicates that there are probably large differences between individual scores.

### **We will be looking at the following measures of spread:**

- range
- quartiles
- interquartile range (IQR)
- variance and
- standard deviation.



# Range

- The range is the difference between the highest and lowest scores in a data set and is the simplest measure of spread.

> We calculate range as:

**Range = maximum value - minimum value**

Example. Data set is {24, 31, 58, 55, 60, 63, 54, 45, 36, 28, 32} marks.

The maximum value is: 63

The minimum value is: 24

The range is 39 (which is  $63 - 24$  )

# Range

- > The maximum value is 24 marks
- > The minimum value is 63 marks
- > The range is 39 marks (which is  $63 - 24 = 39$  marks)

**NOTE: Using the range as a measure of spread can be limited.**

But it does set the boundaries of the scores.

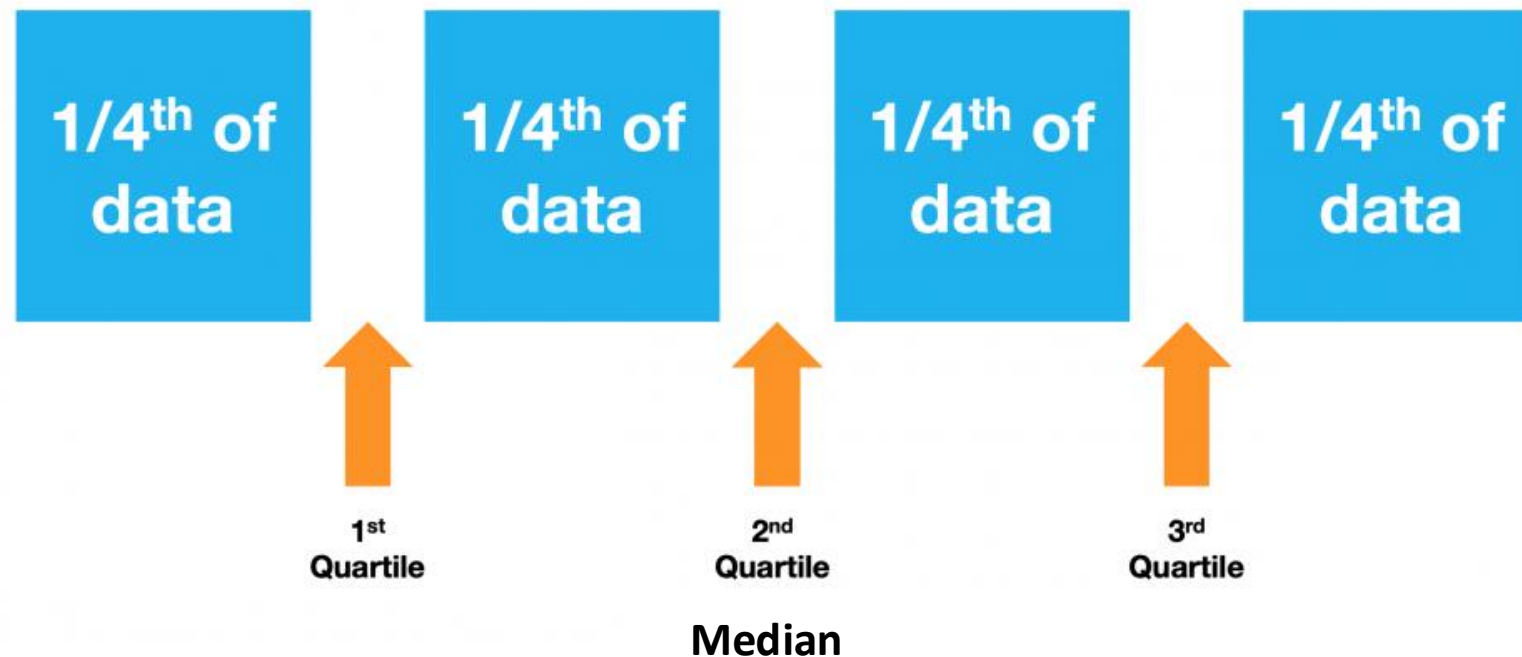
This is useful if you are measuring a variable that has either a critical low or high threshold (or both) that should not be crossed.

For example, if you have recorded the age of school children in your study and your range is 125 years you know you have made a mistake!

# Quartiles

Quartiles tell us about the spread of a data set by breaking the data set into quarters, just like the median breaks it in half.

IMPORTANT: There are three quartiles, namely, Q1, Q2 (median), and Q3.

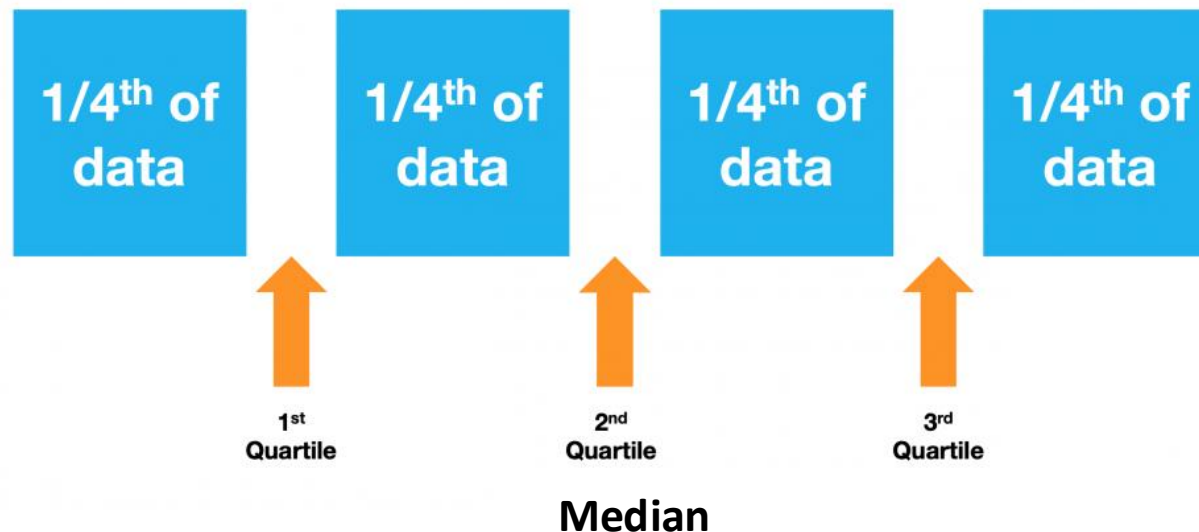


# Quartiles

Example. Data set is {24, 31, 58, 55, 60, 63, 54, 45, 36, 28, 32} marks.

**Step 1.** Find the median (data needs to be sorted ascending order). The median is Q2

**Step 2.** Find the median of each half to obtain Q1 (lower quartile) and Q3 (upper quartile).

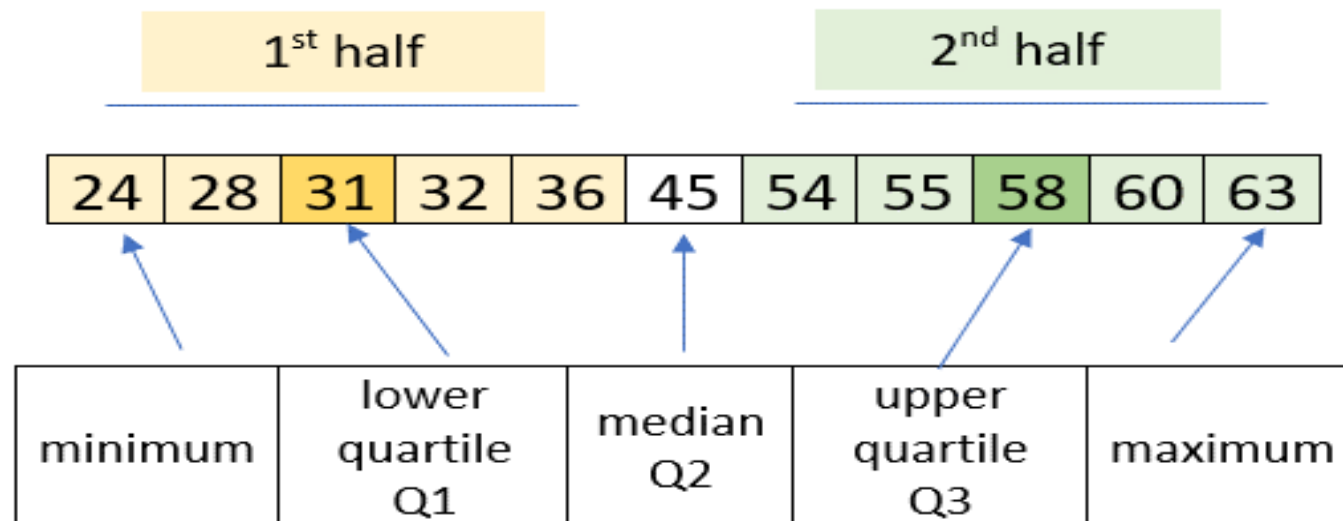


# Quartiles

Example. Data set is {24, 31, 58, 55, 60, 63, 54, 45, 36, 28, 32} marks

**Step 1.** Find the median (data needs to be sorted ascending order). The median is Q2

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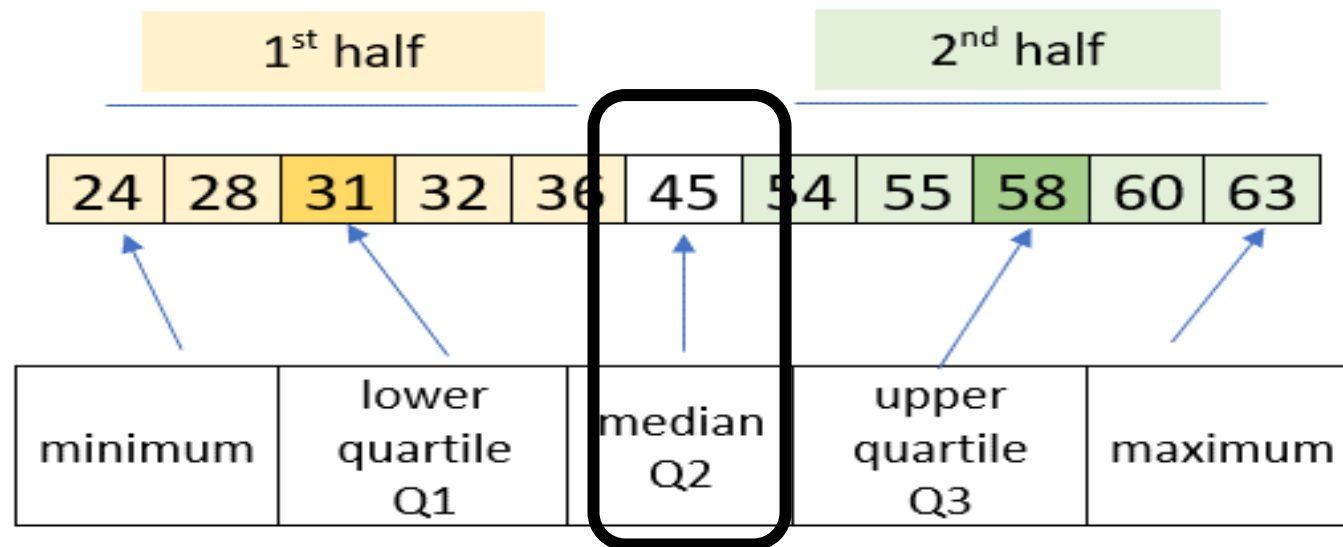


# Quartiles

Interpretation (Final Exam and Labs !)

**I.** The median (Q2) is 45 marks which means that 50% of the students got a mark of 45 or lower

**II (optionally).** The median (Q2) is 45 marks which means that 50% of the students got a mark of 45 or higher

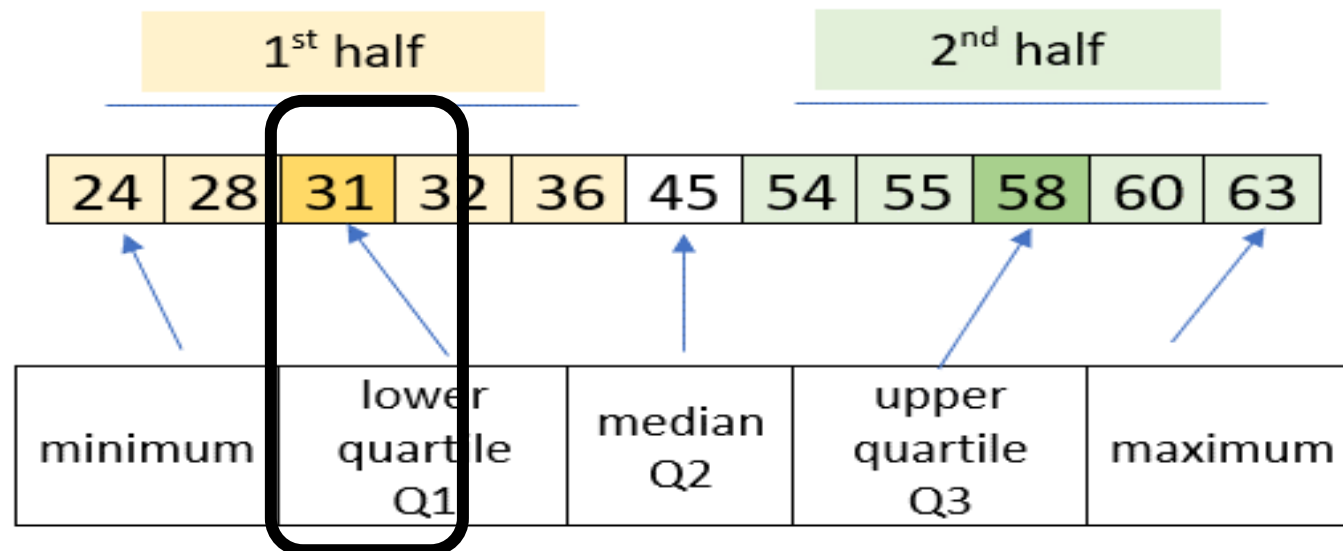


# Quartiles

Interpretation (Final Exam and Labs !)

I. The lower quartile (Q1) is 31 marks which means that 25% of the students got a mark of 31 or lower.

II (**optionally**). The lower quartile (Q1) is 31 marks which means that 75% of the students got a mark of 31 points or higher.



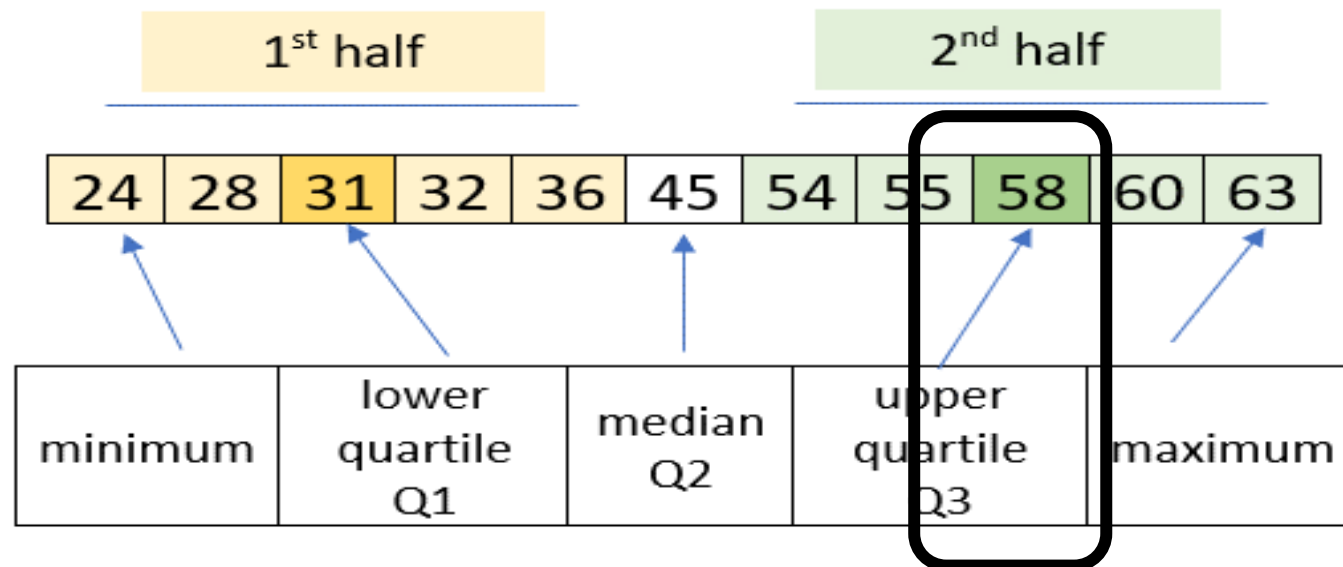


# Quartiles

Interpretation (Final Exam and Labs !)

I. The upper quartile (Q3) is 58 marks which means that 75% of the students got a mark of 58 or lower.

II (**optionally**). The upper quartile (Q3) is 58 marks which means that 25% of the students got a mark of 58 points or higher.



# Interquartile range

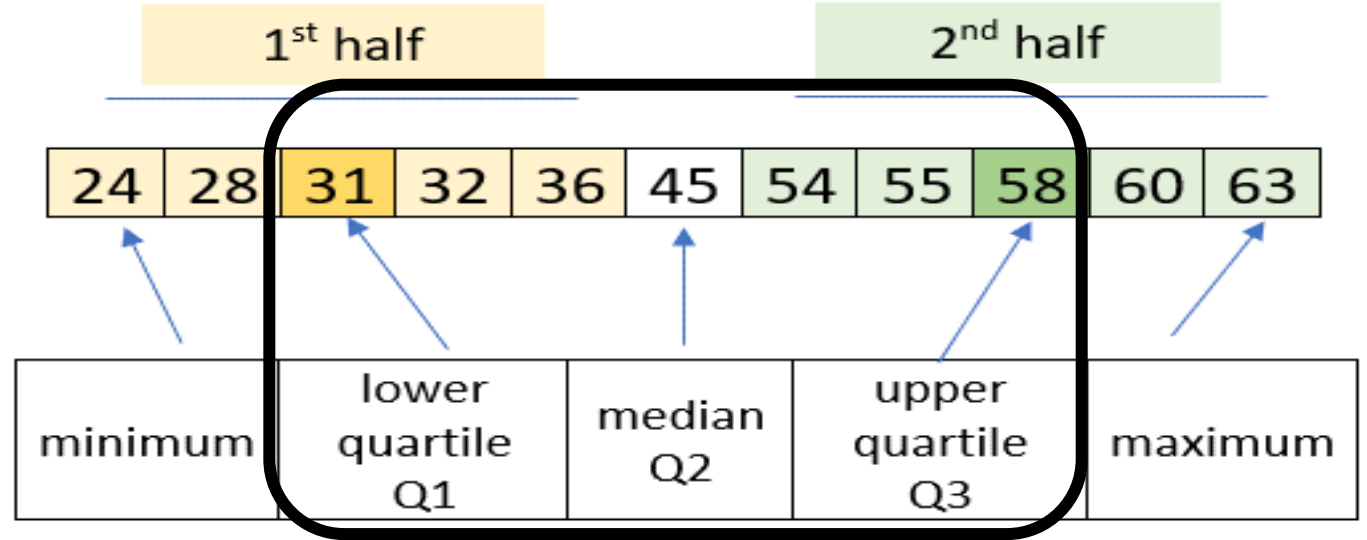
**A common way of expressing quartiles is as an interquartile range.**

**Definition:** The interquartile range (IQR) is given by the difference between the upper quartile (Q3) and the lower quartile (Q1), telling us about the range of the middle half of the scores in the distribution.

For our data set, the IQR is

$$\text{IQR} = 58 - 31 = 27 \text{ marks}$$

> Do not confuse with  
the RANGE



# TAKE AWAY MESSAGES

1. Quartiles are a useful measure of spread because they are much *less affected by outliers* or a *skewed data set* than the equivalent measures of mean and standard deviation.

Example – Data set {1, 3, 3, 5, 7} -> Median is 3, mean = 3.8

Now, if data set {1, 3, 3, 5, 7, 100} -> Median is  $(3 + 5) / 2 = 4$ , mean = 19.8

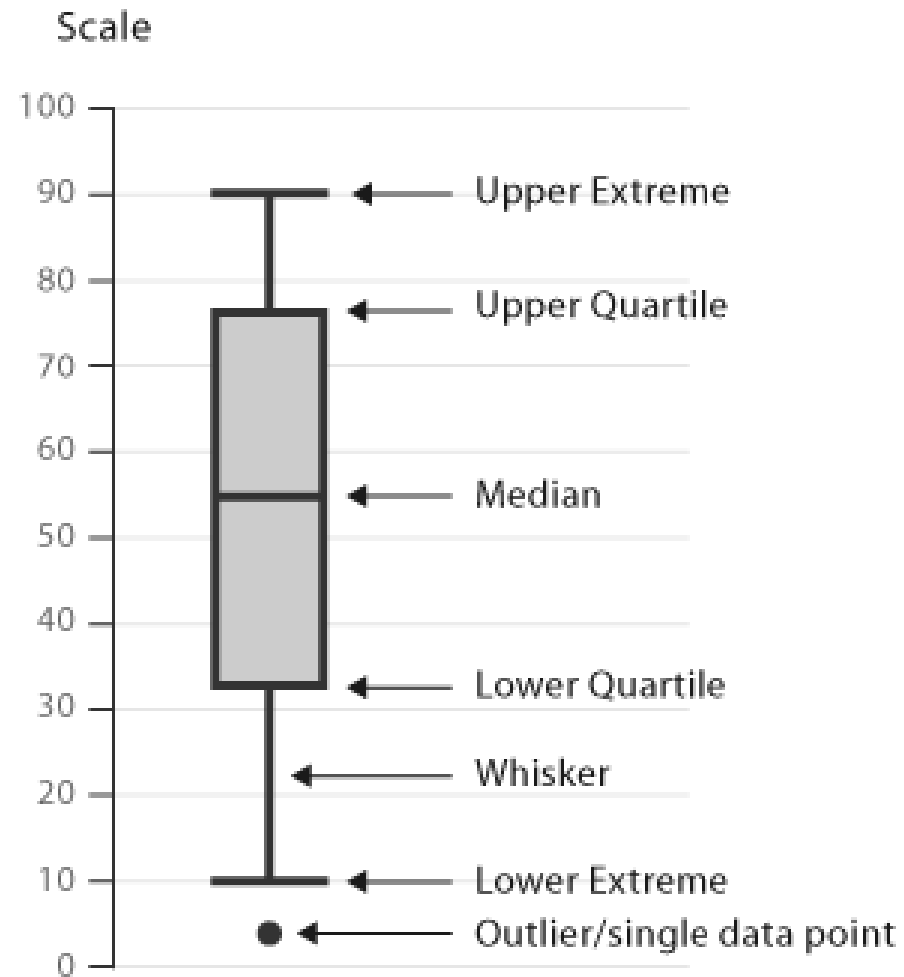
2. Best choice of measure of spread and central tendency when data is skewed or data has outliers: **Median in conjunction with quartiles, interquartile range and range.**

# Important note

Unfortunately there is no universal agreement on the definition of the Quartiles!

- See <https://en.wikipedia.org/wiki/Quartile> for at least 4 different definitions
- Do you see this issue when using Rstudio?

# Box and Whisker graphs

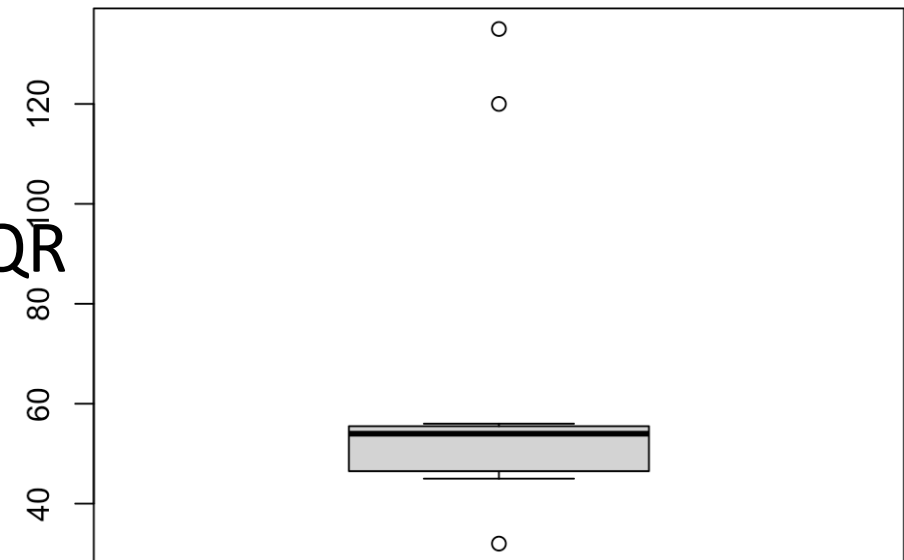


# Outliers

- Outliers are data points that are an abnormal distance from the rest of the data
  - Presence of an outlier may indicate a faulty data point but not always
- ```
>data <- c(32,45,46,47,48,54,55,55,56,120,135)
```

32, 120 and 135 are genuine data points but are flagged as outliers

- Outlier test :  $x > Q3 + 1.5 \times IQR$  or  $x < Q1 - 1.5 \times IQR$



# Boxplots - Example

As production manager, you want to check the performance of 3 filling machines and verify that they produced similar (on average) filling times (given in minutes). The production cost of the Machine 1 is triple the production cost of the other two.

Twenty-four 10,000 litre containers are randomly assigned to the machines which are operated by 3 similarly trained & experienced engineers (so the 'operator' is expected to have no effect on our results). The response is the filling time for each container.

| <u>Machine1</u> | <u>Machine2</u> | <u>Machine3</u> |
|-----------------|-----------------|-----------------|
| 25.40           | 23.40           | 20.00           |
| 26.31           | 21.80           | 22.20           |
| 24.10           | 23.50           | 19.75           |
| 23.74           | 22.75           | 20.60           |
| 25.10           | 21.60           | 20.40           |
| 28.56           | 23.45           | 19.56           |
| 26.19           | 21.56           | 21.89           |
| 23.45           | 20.90           | 21.33           |

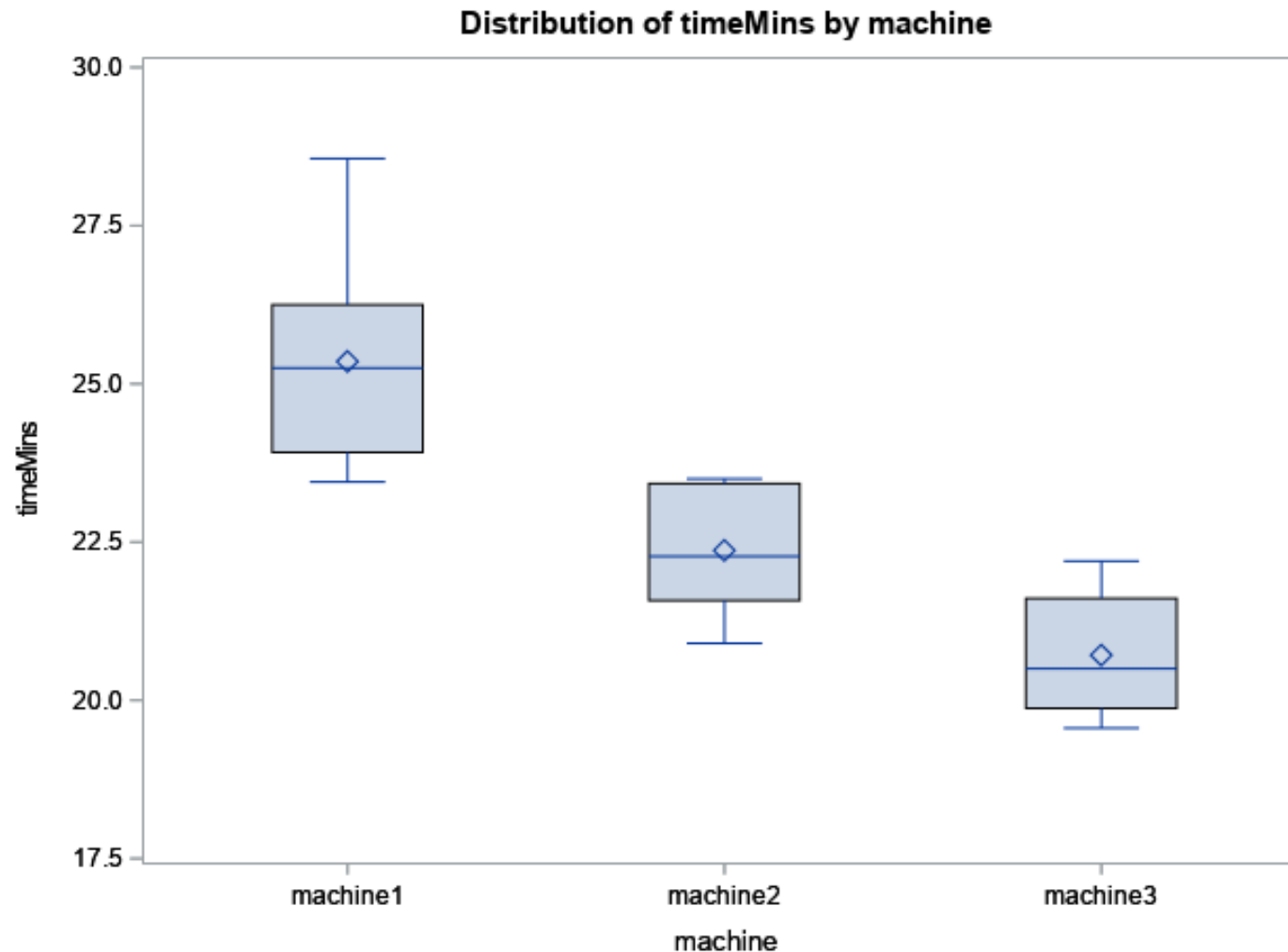
## RESEARCH QUESTIONS:

**Is there evidence indicating that the MEAN filling times differ across the filling machines? Would you suggest to turn off Machine 1? Why?**

NOTE: We have a numeric response (**filling time**) & a factor (ind- variable) as the single criterion- **Machine**.



# 1. PROC BOXPLOT – Example I



From the boxplots, the highest filling time is produced by Machine 1 (28.56 minutes) and the lowest by Machine 3 (19.56 mins), with a difference of 9 minutes (Machine 3 –Machine 1).

There are no outliers: suggesting the data are representative of actual performance rather than some temporary glitch affecting a machine (machine 1 for example!)

Machine 1, appears to yield the highest variability.

Machine 2 fills the containers in 22.37 minutes on average, median time is 22.27 mins

# Summary:

In summary for the salary data set:

**Maximum filling time M1:** 28.56 min

**Minimum filling time M1:** 23.45 min

**Upper quartile M1:** 26.22 min

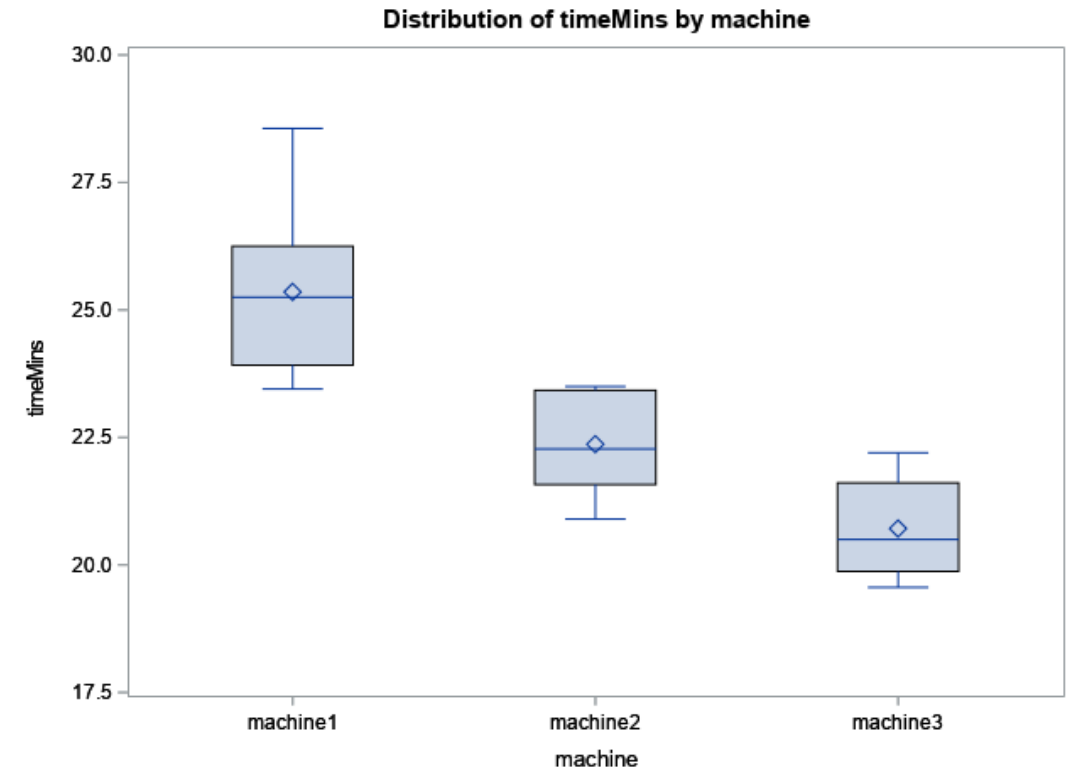
**Median M1:** 25.25 min

**Lower quartile M1:** 24.01 min

**Range M1:**  $28.56 - 23.45 = 5.11$  min

**Interquartile range M1:**  $26.22 - 24.01 = 2.21$  min

**Mean:** 25.36 min



# What's going on in this graph?

<https://www.nytimes.com/2018/01/04/learning/whats-going-on-in-this-graph-jan-9-2018.html>

“box plots showing potential earnings from different degree majors”

What do you notice?

- What this may imply.....

What do you wonder?

- Are there items you noticed that answer what you wonder? Where could you find those answers?

Compare the medians of the different majors (which many of you did), but don't stop there... also compare the other percentiles — 10%, 25%, 75%, and 90%. Which majors tend to have the least career earnings and most career earnings? Do career earnings vary more above or below the median? What does this imply?



# In summary – Today's lecture

<https://bit.ly/3lFmbPO>

<https://bit.ly/3K2ZA9d>

“Numeric variables” and statistical measures



Don't forget that we will be working  
with R and Rstudio this week!



# What did you get out of today?

- What was the purpose of today's lecture?
- What is one key thing you learnt?
- Is there anything that puzzled me?